

Allamp Hardware BOM

Jetson Nano 4GB API-hybrid build - non-tool hardware plus required printed parts

Non-Printed Hardware

Subsystem	Item	Exact part / model	Qty
Core Control	Main controller	NVIDIA Jetson Nano Developer Kit 4GB, MPN 945-13450-0000-100	1
Core Control	System card	SanDisk Ultra microSDXC 64GB UHS-I	1
Motion	Servo	Waveshare ST3215 Servo, 30kg.cm @ 12V, SKU 22414	5
Motion	Servo driver	Waveshare Servo Driver with ESP32, SKU 21593	1
Motion	Servo power	MEAN WELL GST120A12-P1J, 12V 10A 120W	1
Motion	Servo extension	ST / SC serial bus servo extension cable	5
Lighting	LED controller	Raspberry Pi Pico WH	1
Lighting	LED panel	Adafruit NeoPixel NeoMatrix 8x8, 64 RGB LED, Product ID 1487	1
Lighting	LED power	MEAN WELL GST60A05-P1J, 5V 6A 30W	1
Lighting	Logic level shifter	TXS0108E 8-Channel Logic Level Converter Module	1
Lighting	LED resistor	330 ohm 1/4W resistor	5
Lighting	LED capacitor	1000 uF 6.3V or 10V electrolytic capacitor	2
Vision / Audio	Camera	Arducam UB0234, 2MP OV2710 USB2.0 UVC Camera, 1080p, M12 lens	1
Vision / Audio	Microphone array	Seeed Studio ReSpeaker XVF3800 USB 4-Mic Array with Case, Product 6490	1
Vision / Audio	Speaker	Seeed Studio Mono Enclosed Speaker, 4 ohm 5W	1
Power / Wiring	Emergency switch	12V 10A DC inline switch / emergency stop switch	1
Power / Wiring	USB cable	USB-A to USB-C data cable, 0.5m	1
Power / Wiring	USB cable	USB-A to Micro-USB data cable, 0.5m	1
Power / Wiring	USB extension	USB-A extension cable, 0.3m or 0.5m	1
Power / Wiring	DC barrel terminal	5.5mm x 2.1mm DC barrel screw terminal adapter	4
Power / Wiring	Power connector	WAGO 221-413 lever connector	10
Power / Wiring	Power wire	22AWG red silicone wire	2m
Power / Wiring	Power wire	22AWG black silicone wire	2m
Power / Wiring	Signal wire	24AWG silicone wire	2m

3D Printed Parts

Group	File	Purpose	Qty
Original LeLamp 3D Prints	LampBase.3mf	Base structure	1
Original LeLamp 3D Prints	LampBase - Cover.3mf	Base cover	1
Original LeLamp 3D Prints	LampArm (Base-Elbow).3mf	Base-to-elbow arm	1
Original LeLamp 3D Prints	LampArm (Elbow-Wrist).3mf	Elbow-to-wrist arm	1
Original LeLamp 3D Prints	LampArm (Pitch).3mf	Pitch joint part	2
Original LeLamp 3D Prints	LampHead.3mf	Lamp head shell	1
Original LeLamp 3D Prints	LampHead - Diffuser.3mf	Lamp diffuser	1
AllLamp Adapter Kit	AllLamp_Jetson_Nano_Base_Tray.3mf	Loose external tray for Jetson Nano	1
AllLamp Adapter Kit	AllLamp_Electronics_Side_Deck.3mf	Servo driver and Pico WH side deck	1
AllLamp Adapter Kit	AllLamp_Head_Camera_Mount.3mf	Reversible Arducam UB0234 head mount	1
AllLamp Adapter Kit	AllLamp_NeoMatrix_Holder.3mf	NeoMatrix holder behind diffuser	1
AllLamp Adapter Kit	AllLamp_ReSpeaker_External_Mount.3mf	External ReSpeaker XVF3800 mount	1
AllLamp Adapter Kit	AllLamp_Cable_Clip_6mm.3mf	USB and signal cable routing	2-4
AllLamp Adapter Kit	AllLamp_Cable_Clip_10mm.3mf	Power and servo cable routing	2-4

Notes: This BOM follows the basic Jetson Nano 4GB profile. NVMe storage is not required. The AllLamp adapter kit is designed as reversible loose-fit hardware; do not cut the original LeLamp printed files for the first prototype.